Method and System for High-Radix Data Representation Using Parametric Mathematical Encoding

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Title: Method and System for High-Radix Data Representation Using Parametric Mathematical Encoding

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Cross-Reference to Related Applications

This application claims the benefit of and priority to the concepts and systems described in U.S. Patent Applications for "High-Radix-Number-Representation", "High-Precision-Fractional-Number-Representation", and "Ultra-Wide-Parametric-Bus-Architecture", the contents of which are hereby incorporated by reference in their entirety.

Abstract

A method and system for representing and transmitting numerical data across a communication channel, wherein the data is encoded as a sequence of parametric components. These components define not only numerical values but also the mathematical operations to be performed upon them. This parametric, mathematical encoding allows for the transmission of numbers with effectively unlimited precision and magnitude, circumventing the physical limitations of the communication channel's radix. The receiving system interprets this sequence,

or n-gram, as a 'recipe' to reconstruct the original number, enabling the representation of vast integers and non-truncating fractions.

Background of the Invention

Traditional digital communication systems represent numbers in a binary format, where the precision and maximum value are constrained by the number of bits (the bus width) available in a single transmission cycle. While wider buses and higher-radix systems can increase these limits, they are ultimately bound by the physical characteristics of the transmission medium and the transceivers. Representing very large integers or high-precision fractional numbers requires multiple transmission cycles or complex data structures, introducing latency and overhead. Existing methods do not provide a mechanism to transmit the conceptual meaning of a number separately from its truncated, channel-limited representation.

Summary of the Invention

The present invention provides a method and system that encodes numbers not as a single, fixed-radix value, but as a parametric sequence of values and mathematical operators. This sequence, referred to as an n-gram, is transmitted over a communication channel, such as a parallel optical bus. The n-gram is composed of multiple symbols, where each symbol can represent an operand (an integer or fractional value) or a mathematical operator (e.g., add, multiply, divide, exponentiate).

Upon reception, the system parses the n-gram, interpreting it as a set of instructions. It applies the encoded operators to the encoded operands in a predefined order to reconstruct the intended numerical value. For example, a fractional number like 7/16 can be transmitted not as its floating-point approximation, but as the sequence of symbols representing '7', 'divide', and '16'. The receiving system can then calculate the result to any desired level of precision. Similarly, an integer far exceeding the channel's maximum single-word value (MAX_INT) can be represented as a product or exponentiation of smaller, transmissible values (e.g., MAX_INT * MAX_INT). This method effectively trades a small amount of channel parallelism for a geometric expansion of the representable number space.

Statement of License Grant to the United States Federal Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the method of encoding, transmitting, and decoding a number using parametric mathematical encoding.

FIG. 2 is a conceptual diagram illustrating the encoding and transmission of a high-precision fractional number.

FIG. 3 is a conceptual diagram illustrating the encoding and transmission of an integer exceeding the native maximum value of the communication channel.

Detailed Description of the Invention

Referring to **FIG. 1**, the flowchart (100) shows the overall process. First, a number to be transmitted is identified (110). The system then decomposes this number into a sequence of operands and mathematical operators (120). This sequence, or n-gram, is then encoded into a set of symbols compatible with the transmission channel (130), for instance, assigning specific wavelengths of light to represent numbers and operators like 'add' or 'multiply'. The encoded n-gram is then transmitted over the channel (140). A receiving system captures the n-gram (150) and decodes the symbols back into their constituent operands and operators (160). Finally, the receiver executes the mathematical operations on the operands to reconstruct the original numerical value to its desired precision (170).

FIG. 2 provides a conceptual illustration (200) of encoding a fractional number. The fraction 7/16 (210) is decomposed into the sequence: operand '7', operator 'divide', and operand '16'. These are encoded into distinct symbols (220) - for example, different colors or wavelengths of light - and transmitted as an n-gram (230) across a parallel channel. The receiver interprets

this "recipe" to reconstruct the fraction 7/16, avoiding any truncation or precision loss during transit.

FIG. 3 shows the encoding of a very large integer (300). The number to be sent exceeds the maximum integer value (MAX_INT) representable by a single symbol on the channel. The number is decomposed into a formula, such as MAX_INT raised to the power of 2 (310). This is encoded into the sequence: operand 'MAX_INT', operator 'exponentiate', operand '2'. This n-gram of symbols (320) is transmitted, allowing the receiver to calculate a number far larger than the channel could otherwise carry in a single, parallel transmission. This demonstrates the trade-off of using multiple symbols (pixels in an optical frame) to achieve a much higher effective radix.

Claims

What is claimed is:

1. A method for transmitting numerical data, comprising:
    - decomposing a numerical value into a sequence of one or more operands and one or more mathematical operators;
    - encoding said sequence into an n-gram of symbols, wherein each symbol corresponds to an operand or an operator;
    - transmitting the n-gram over a communication channel;
    - receiving the n-gram at a destination;
    - decoding the n-gram to retrieve the sequence of operands and operators; and
    - applying the operators to the operands to reconstruct the original numerical value.
2. The method of claim 1, wherein the communication channel is a parallel optical bus.
3. The method of claim 2, wherein the symbols are encoded as distinct wavelengths of light.
4. The method of claim 1, wherein the numerical value is a fractional number, and the reconstructed value retains the full precision of the original fraction.
5. The method of claim 1, wherein the numerical value is an integer larger than the maximum value representable by a single symbol on the communication channel.

6. A system for transmitting numerical data, comprising:

    - an encoder configured to decompose a numerical value into a sequence of operands and mathematical operators and encode said sequence into an n-gram of symbols;
    - a transmitter for sending the n-gram over a communication channel;
    - a receiver for capturing the n-gram from the channel; and
    - a decoder configured to decode the n-gram and apply the operators to the operands to reconstruct the numerical value.

7. The system of claim 6, wherein the communication channel is a parallel optical data bus and the symbols are transmitted as parallel wavelengths of light.

8. The system of claim 6, wherein the decoder is configured to calculate a fractional number from the decoded n-gram to a user-specified level of precision.

9. The system of claim 6, wherein the n-gram represents an integer value greater than the native radix of the communication channel.

Drawings

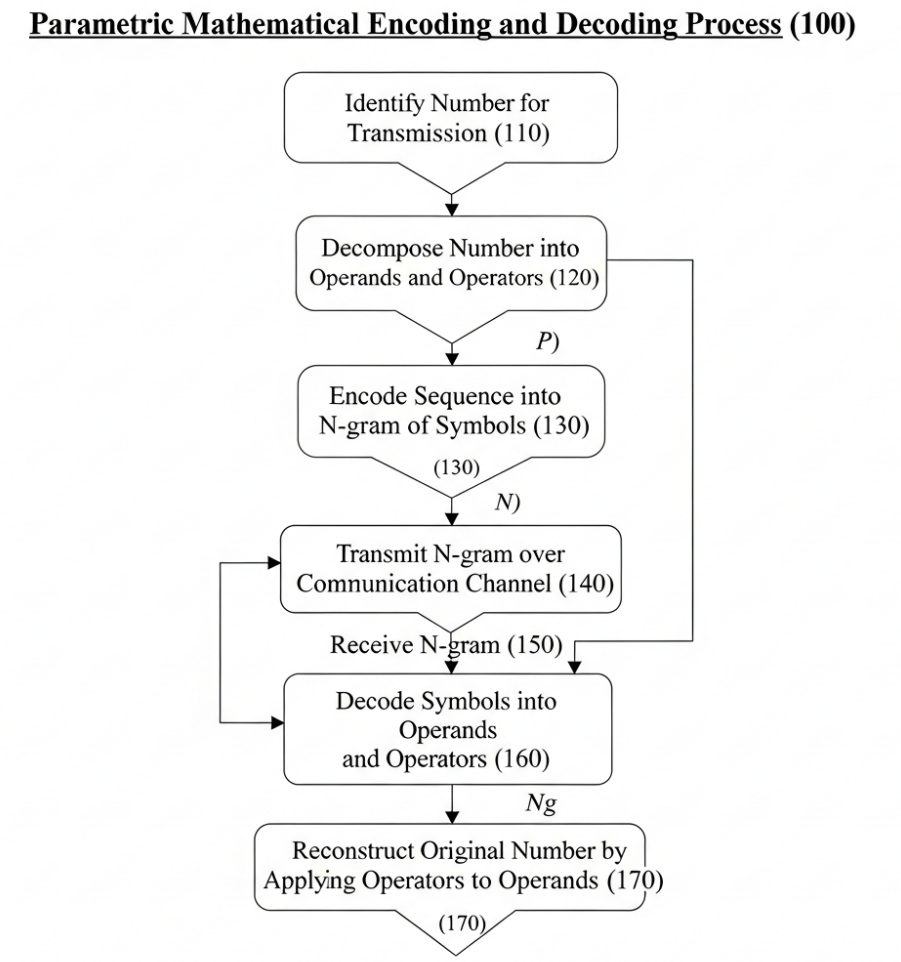


Figure 1: Flowchart illustrating the method of encoding, transmitting, and decoding a number using parametric mathematical encoding.

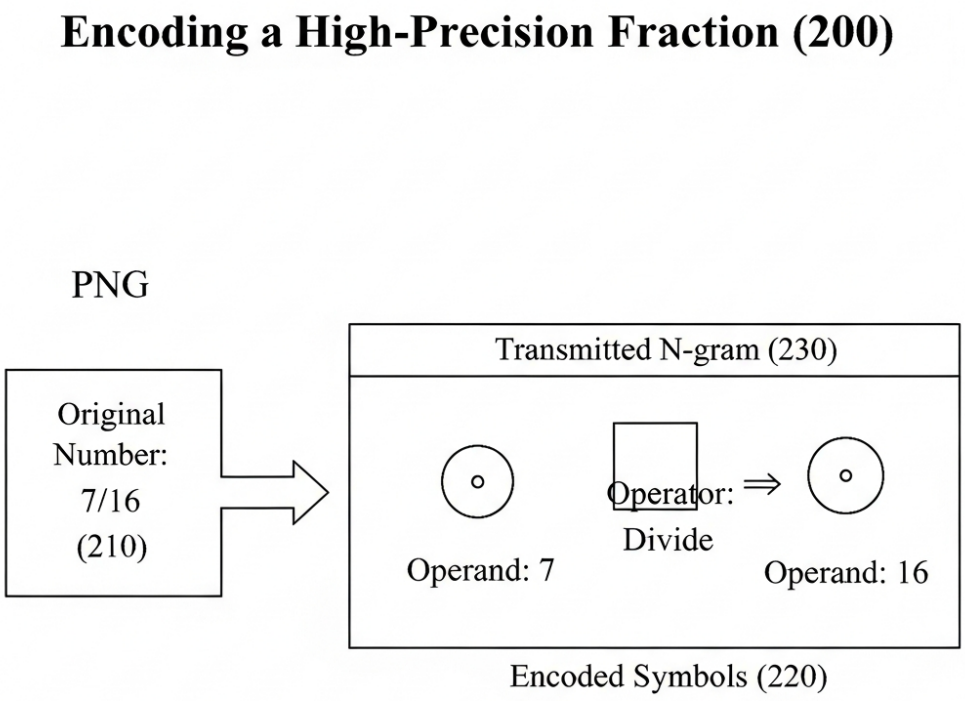


Figure 2: Conceptual diagram illustrating the encoding and transmission of a high-precision fractional number.

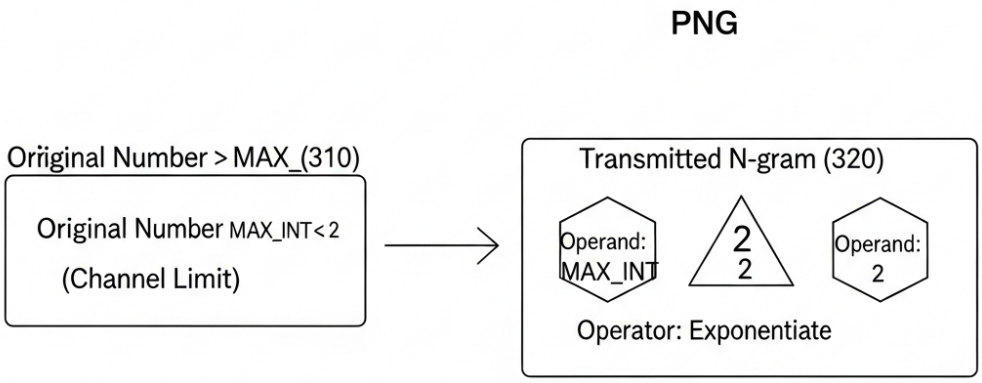


Figure 3: Conceptual diagram illustrating the encoding and transmission of an integer exceeding the native maximum value of the communication channel.